BEU CONNECT

Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2019

Time: 03 Hours

Code: 031508 ***Subject: POWER SYSTEM-II*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Write short answers of the following (Attempt any seven): **[14]**

(a) Describe phase shift in Star-delta Transformers.

(b) Define voltage unbalance.

(c) Explain the term 'voltage stability'.

(d) What is 'equal area criterion'?

(e) Differentiate between steady state stability and transient stability of power system.

(f) What for surge tank is provided?

(g) What are the advantages of using per unit system?

(h) Define stability limit of power system.

(i) What is sub-station? Name the factors that should be taken care of while designing and erecting a sub-station.

(j) What is reheater?

Q.2 Solve both questions : **[14]**

(a) What are the functions of economizer and superheater in a thermal power plant?

(b) A thermal power plant spends rupees 25 lakhs in one year for coal consumption. The coal has heating value of 5000 kcal/kg and costs rupees 500 per ton. If the thermal efficiency is 35% and electrical efficiency is 90%, find the average load on the power plant.

Q.3 Solve both questions : **[14]**

(a) What is sub-station? Name the factors that should be taken care of while designing and erecting a sub-station.

(b) Give a general layout of a pumped hydro plant. Explain the function of different components in plant.

Q.4 Solve both questions : **[14]**

(a) A 500 MVA, 11 kV, 3 phase alternator was subjected to different types of faults. The fault currents are as given below: Three phase fault - 2000 A; Line to line fault - 2600 A; Line to ground fault - 4200 A. The generator neutral is solidly grounded. Find the values of the sequence reactances of the alternator. Assume that generator has negligible winding resistance.

(b) What do you understand by positive, negative and zero sequence impedances? Discuss them with reference to synchronous generators, transformers and transmission lines.

Q.5 Solve this question : **[14]**

The voltages across a 3-phase unbalanced load are V_a= 300 V, V_b= $300\angle -90^\circ$V and V_c= $800\angle -143.1^\circ$V, respectively. Determine the sequence components of voltages. Phase sequence is abc.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

BEU CONNECT

Q.6 Solve both questions : **[14]**

(a) Define the following with example in connection with a power supply system: (i) Connected load (ii) Maximum demand (iii) Demand factor (iv) Load factor.

(b) A generating station has a connected load of 450 MW and a maximum demand of 250 MW; the units generated being 615×10^6 per annum. Calculate the demand factor and load factor.

Q.7 Solve both questions : **[14]**

(a) Explain the term sequence impedance for transmission lines with example and mathematical expressions.

(b) Prove that a line-to-ground fault at the terminals for a synchronous generator with solidly grounded neutral is more severe than a three-phase fault.

Q.8 Solve this question : **[14]**

What is a 3-phase unsymmetrical fault? Discuss the different types of symmetrical fault that can occur on? Obtain an expression for critical angle with the aid of equal area criterion.

Q.9 Solve both questions : **[14]**

(a) Derive the swing equation of synchronous machine swinging against an infinite bus. Clearly state the assumption in deducing the swing equation.

(b) Distinguish between steady state, transient and dynamic stability. What do you mean by critical clearing angle?

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store